

Mathematics

Mathematics is the abstract science of numbers, quantity, and space, both as abstract concepts (pure mathematics) and as they are applied to other disciplines such as physics and engineering (applied mathematics). It involves the study of structures, patterns, relationships, and changes, using logical reasoning, symbolic representation, and computational techniques to solve problems and understand the world around us.

- When we are dealing with mathematics on large scale than it is called data science.

Why is mathematics for data science?

1. Count (numbers e.g. money in Rs, usd, pounds and euro)
2. Measurements (unit. Scale)
3. Logical reasoning
4. Rigorous proof
5. Relations

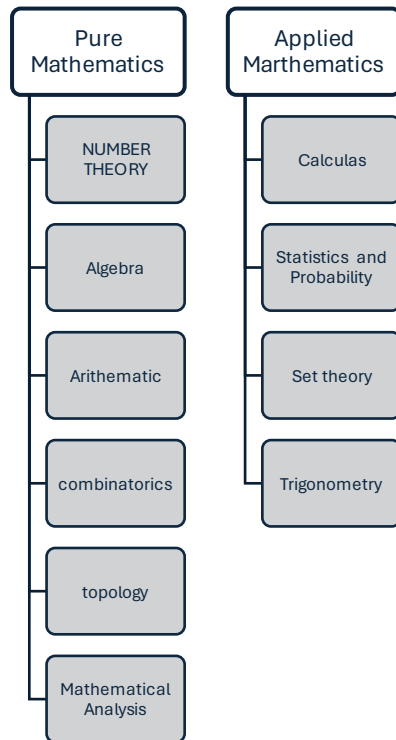
Write the five different definitions of mathematics by five different scientists?

1. **Carl Friedrich Gauss:** Mathematics is the queen of the sciences, a field that studies quantities, structures, and spaces through logical reasoning and systematic calculations.
2. **Albert Einstein:** Mathematics is a tool for understanding the physical world, a precise language that helps us describe the laws of nature.
3. **Bertrand Russell:** Mathematics is a subject in which we never know what we are talking about, nor whether what we are saying is true, highlighting its abstract nature and logical structure.
4. **David Hilbert:** Mathematics is a game played according to certain simple rules with meaningless marks on paper, emphasizing the formal and symbolic nature of mathematical practice.
5. **Leonhard Euler:** Mathematics is the science of quantity and space, involving the study of numbers and shapes, and the relationships between them through formulas and equations.

Branches of mathematics

- Number theory (Number theory is a branch of mathematics that studies numbers and the relationships between them, especially focusing on whole numbers and their properties.)
- Linear algebra (Linear algebra is a branch of mathematics that deals with vectors, matrices, and the operations that can be performed on them to solve systems of linear equations and understand spaces.)
- Statistics (Statistics is a branch of mathematics that involves collecting, analyzing, interpreting, and presenting data to understand patterns and make informed decisions.)
- Probability (Probability is the branch of mathematics that measures how likely it is for something to happen.)
- Mathematical physics (Mathematical physics is a field that uses mathematical methods to solve problems and understand concepts in physics.)
- Calculus and analytical geometry (Calculus is a branch of mathematics that studies how things change and includes concepts like differentiation and integration. Analytical geometry, also known as coordinate geometry, is the study of shapes and figures using a coordinate system to represent them with algebraic equations.)

Types of mathematics



Are statistics the branch of mathematics or not?

Yes, statistics is a branch of mathematics. It deals with collecting, analyzing, interpreting, presenting, and organizing data.

Number Theory

Number theory is a branch of mathematics that focuses on studying whole numbers and their properties, patterns, and relationships. They are sometimes called higher arithmetic.

NUMBERS AND THEIR CLASSES

1. **Natural Numbers (Counting Numbers):** Positive integers starting from 1.

Example: 1, 2, 3, 4, ...

2. **Whole Numbers:** Natural numbers including 0.

Example: 0, 1, 2, 3, 4, ...

3. **Integers:** Whole numbers and their negative counterparts.

Example: -3, -2, -1, 0, 1, 2, 3, ...

4. **Rational Numbers:** Numbers that can be expressed as the quotient of two integers (fractions).

Example: $\frac{1}{2}$, $-\frac{3}{4}$, 5, 0.75

6. **Irrational Numbers:** Numbers that cannot be expressed as simple fractions; their decimal expansions are non-repeating and non-terminating.

Example: $\sqrt{2}$, π , e

6. **Real Numbers:** All rational and irrational numbers.

Example: -2, 0, 1.5, $\sqrt{3}$, π

7. **Complex Numbers:** Numbers that have a real part and an imaginary part.

Example: $3 + 4i$, $-2 - i$, $0 + 2i$

8. **Prime Numbers:** Natural numbers greater than 1 that have no positive divisors other than 1 and themselves.

Example: 2, 3, 5, 7, 11

9. **Composite Numbers:** Natural numbers greater than 1 that are not prime; they have more than two positive divisors.

Example: 4, 6, 8, 9, 12

10. **Even Numbers:** Integers divisible by 2.

Example: -4, 0, 2, 6, 10

11. **Odd Numbers:** Integers not divisible by 2.

Example: -3, 1, 5, 7, 11

12. **Triangular Numbers:** Numbers that can form an equilateral triangle. They are the sum of the first n natural numbers.

Example: The 3rd triangular number is 6 because $1 + 2 + 3 = 6$. The first few triangular numbers are 1, 3, 6, 10, 15, ...

13. **Perfect Numbers:** Numbers that are equal to the sum of their proper divisors (excluding the number itself).

Example: 6 is a perfect number because its divisors are 1, 2, and 3, and $1 + 2 + 3 = 6$. The first few perfect numbers are 6, 28, 496, ...

14. **Fibonacci Numbers:** A sequence of numbers where each number is the sum of the two preceding ones, starting from 0 and 1.

Example: The sequence starts as 0, 1, 1, 2, 3, 5, 8, 13, ... Here, $0 + 1 = 1$, $1 + 1 = 2$, $1 + 2 = 3$, $2 + 3 = 5$, and so on.

Each number in sequence is Fibonacci number. They appear frequently in nature and have many properties and applications in mathematics and science. They create a kind of line or relation.

Number theory, often termed as the queen of mathematics primarily revolves around investigating the inherited properties of numbers, their relationship and structure they form. It is a realm that explores the behavior and characteristics of members of such a prime and their role in various mathematical phenomena.

➤ **An important point to note is that the set of numbers is called data. Data science is dealing with big data.**

APPLICATIONS OF NUMBER THEORY

Number theory has several practical applications in various fields. Here are some simple explanations:

1. **Cryptography:** Number theory is fundamental to cryptography, the science of secure communication. Techniques like RSA encryption relies on the difficulty of factoring large numbers, which is a problem studied in number theory.

2. **Computer Science:** Number theory concepts are used in computer science for developing algorithms and data structures. For instance, prime numbers play a crucial role in certain algorithms like hashing and primality testing.

3. **Coding Theory:** Number theory is applied in coding theory, which deals with the efficient transmission of data over noisy channels. Concepts like error-correcting codes and checksums are based on number theory principles.

4. **Finance:** Number theory is used in finance for tasks like generating secure random numbers for cryptographic protocols, analyzing market trends using mathematical models, and ensuring the security of financial transactions.

5. **Physics and Engineering:** Number theory concepts are employed in various branches of physics and engineering for modeling physical phenomena, analyzing data, and solving complex equations.

6. **Number Systems and Representations:** Understanding number theory helps in developing efficient number systems and representations used in computer science, such as binary, hexadecimal, and octal systems.

7. **Data Integrity:** It refers to the accuracy and reliability of data throughout its lifecycle. It ensures that the data remains unchanged and consistent, maintaining its intended meaning and value. Essentially, data integrity ensures that the information stored or transmitted is complete, accurate, and secure, without any unauthorized alterations or corruption.

8. **Big data:** It refers to large volumes of data that are too complex or extensive for traditional data processing methods to handle effectively. This data is characterized by its volume, velocity, and variety, meaning it's massive in size, generated rapidly, and comes in various formats like text, images, videos, and sensor data. Big data analysis involves using specialized tools and techniques to extract valuable insights and patterns from this vast amount of information, which can be used for decision-making, forecasting, and improving processes across various industries and sectors.

Pros of learning Number theory

- ❖ Problem solver
 - Logic built.
 - Analytical ability.
 - Data scientist.
- ❖ Optimization.
- ❖ Unlock new algorithms.
- ❖ Competitive edge.
 - Mathematics
 - Logical analysis.
 - Theoretical.
 - Pattern.
 - Complex.

CORE CONCEPTS OF NUMBER THEORY

Factors

Factors are the numbers that can exactly divide a number without leaving a remainder.

For example, 12 is a number whose factors are 1,2,3,4,6,12.

Multiples

Multiples are what you get when you multiply a number by an integer. **For example, the first five multiple of 4 are 4,8,12,16 and 20.**

DIVISIBLE RULE IN MATHEMATICS

Let us have a number that is 424262 respectively.

- **Rule 2:**

When last is the even number.

- **Rule 3:**

A number that is divisible by 3 if this sum is divisible by 3.

For example, we have a number that is 123 respectively. $1+2+3=6$, when divided by 3 we have result that is 2.

- **Rule 4:**

A number is divisible by 4 if its last two digits are divisible by 4.

For example, 4844 and 12359124 both are divisible by 4 because their last 2 digits of numbers are divisible by 4.

- **Rule 5:**

A number is divisible by 5 if the number has a last digit, that is 5 or 0.

For example, the number is 525 or 5050 both the number have the last digit that is 5 or 0 so divisible by 5.

- **Rule 6:**

If a number is following the rule 2 and rule 3 then it must be following the rule 6.

- **Rule 9:**

A number whose digit sum is divisible by 9 is following the rule 9.

For example, we have a number that is 9819, the sum of the number is $9+8+1+9$ is 27 and divisible by 9 we get 3.

- **Rule 10:**

A number whose unit digit is 0 follows the rule of 10.

For example, we have numbers 5050 and 5020.

LCM AND GCD

Sure, let's break it down:

1. LCM (Least Common Multiple): The LCM of two or more numbers is the smallest multiple that is divisible by each of the numbers.

Example: Let's find the LCM of 4 and 6. The multiples of 4 are 4, 8, 12, 16, ... and the multiples of 6 are 6, 12, 18, 24, ... The smallest number that appears in both lists is 12. So, the LCM of 4 and 6 is 12.

2. GCD (Greatest Common Divisor): The GCD of two or more numbers is the largest number that divides each of them without leaving a remainder.

Example: Let's find the GCD of 12 and 18. The divisors of 12 are 1, 2, 3, 4, 6, 12, and the divisors of 18 are 1, 2, 3, 6, 9, 18. The largest number that appears in both lists is 6. So, the GCD of 12 and 18 is 6.

How to find the lcm and gcd in mathematics?

To find the LCM and GCD of numbers, you can use various methods:

Prime Factorization Method: Decompose each number into prime factors and then find the product of the highest powers of all prime factors for LCM and find the product of the common prime factors for GCD.

Listing Multiples or Divisors: List the multiples for LCM and common divisors for GCD, and then find the least or greatest accordingly.

Euclidean Algorithm: A fast and efficient method for finding GCD by repeatedly applying the division algorithm.

MODULER ARITHMETIC

It is often termed as the arithmetic of the clock that introduces a unique way to handle numbers. It is a system where in reaching a certain value known as the module. This concept has widespread applications from computer science to cryptography and lays the groundwork for many advanced mathematical theories.

Modulus

$7 \bmod 5 = 7/5$ remainder is 2.

At the heart of modular arithmetic is a modulus a specific number that sets the threshold for wrapping around. It acts like a reset point beyond which numbers revert to set range.

$15 \bmod 4 = 15 / 4$ with remainder 3.

$100 \bmod 10 = 100/10$ with remainder zero.

ASSIGNMENT

1. What is modulus?
2. How to calculate the modulus?
3. Mathematical equations of modulus?
4. How to find it for different numbers?

Answer

Modulus is the remainder after division. To calculate it, use the modulo operator (%). For example, $7 \bmod 3 = 17 \bmod 3 = 17 \bmod 3 = 1$. It's like finding what's left after sharing items equally among a group. So, for different numbers, divide them and see what's left over. For instance, $10 \bmod 4 = 2$ and $210 \bmod 4 = 2$.

Algebraic numbers

Numbers that are roots of a non-zero polynomial equation with integer coefficients.

ALGEBRA

A branch of mathematics that uses symbols and letters to represent the number of quantities in formula or equations.

It is like a puzzle where we solve for unknown value and make equations true.

For example, if we have $5 + ? = 10$. Algebra helps to figure missing value as 5.

The keywords we used are:

- Arithmetic (+, -, *, /)
- Variables ($5 + x = 10$, find x here x is a variable)

- Expression (**$3x+2$ here $+$ is operator, x is variable and 2 is a number. As expression that contains number variable and operators is called mathematical expression.**)
- Equation (**$3x+2y=5+3z$**)
- Solving the equation

WHY IS ALGEBRA IS IMPORTANT?

1. Problem solving (**equipped the learner with a skill to solve a variety of real word problem**)
2. Critical thinking (**enhance logical reasoning and critical thinking by presenting problem in a good manner**)
3. Foundation of advance studies (**mastery of algebra is essential for advance studies in math, science engineering and other fields**)

How to learn algebra?

1. Understand basics (**grasps the fundamental operation and principal like $+$, $-$, $*$, $/$.**)
2. Practice regularly (**consistent practice is a key to master algebra engage in solving different algebraic equations to improve skills**).
3. Seek help (**do not hesitate to seek help when needed or you are stuck**).

Application of algebra

1. **Matrix operations:** Used in computer graphics for transformations and in solving systems of equations in engineering and physics.
2. **Eigenvalues and eigenvectors:** Essential in quantum mechanics for understanding particle behavior and in engineering for stability analysis of systems.
3. **Linear algebra:** Forms the backbone of various mathematical models and algorithms used in fields like economics, physics, and engineering.
4. **Optimization:** Applies algebraic techniques to find the best solutions to problems, such as maximizing profits in business or minimizing costs in manufacturing.
5. **Solving systems of equations:** Crucial in engineering for designing structures, in economics for analyzing markets, and in physics for modeling physical phenomena.
6. **Machine learning algorithms:** Utilize linear algebra for tasks like feature extraction, dimensionality reduction, and parameter optimization in predictive modeling.
7. **Deep learning algorithms:** Rely on linear algebra for neural network computations, including matrix multiplications, activations, and backpropagation.
8. **Graph theory:** Uses algebraic methods to analyze and model relationships between objects, applicable in social networks, transportation systems, and computer networks.
9. **Natural language processing:** Employs linear algebra for tasks like word embeddings, semantic analysis, and sentiment classification in understanding and processing human language.
10. **Image recognition:** Relies on linear algebra for image processing techniques such as convolution, feature extraction, and pattern recognition.
11. **Clustering and classification:** Utilizes algebraic methods for grouping data points into clusters and categorizing them into classes, used in various fields like marketing, healthcare, and finance.

Applications of algebra in ai and data science

1. **Feature Engineering:** Algebra helps in creating new features from existing data, like combining variables or transforming them, to improve the performance of machine learning models.
2. **Linear Regression:** Algebra is used to find the best-fit line that describes the relationship between input features and output labels in regression problems.
3. **Classification:** Algebraic techniques are applied in algorithms like logistic regression and support vector machines to classify data into different categories based on input features.
4. **Clustering:** Algebra helps in grouping similar data points together in unsupervised learning algorithms like k-means clustering by minimizing distances between points
5. **Dimensionality Reduction:** Techniques like principal component analysis (PCA) use algebra to reduce the number of input features while preserving important information, making data easier to visualize and analyze.
6. **Neural Networks:** Algebra is fundamental in the design and training of neural networks, where matrix operations are extensively used for computations between layers of neurons.
7. **Natural Language Processing (NLP):** Algebra is applied in tasks like word embeddings, where words are represented as vectors in a high-dimensional space, enabling machines to understand and process language.
8. **Image Processing:** Algebraic operations are used to manipulate and analyze images, such as convolutional neural networks (CNNs) applying filters to detect features in images.
9. **Recommendation Systems:** Algebra helps in building recommendation systems by finding similarities between users or items based on their features or interactions.
10. **Optimization:** Algebraic optimization techniques are used to fine-tune machine learning models, adjusting parameters to minimize errors and improve performance.

Types of algebra

1. **Pre-algebra:** Introduces basic concepts like arithmetic operations, fractions, and integers, preparing students for more advanced algebraic topics.
2. **Elementary algebra:** Covers fundamental algebraic operations, including solving equations, working with variables, and understanding basic algebraic expressions and properties.
3. **Intermediate algebra:** Builds upon elementary algebra, covering more advanced topics such as quadratic equations, polynomials, exponents, and functions.
4. **Linear algebra:** Focuses on the study of vectors, matrices, and linear transformations, used extensively in fields like physics, engineering, and computer science.
5. **Modern algebra:** Explores abstract algebraic structures such as groups, rings, and fields, emphasizing theoretical concepts rather than concrete computations.
6. **Boolean algebra:** Deals with logical operations and expressions using variables that can only take on two values (true or false), foundational in digital electronics and computer science.
7. **Relational algebra:** A mathematical system used in database theory to manipulate and query relational databases using algebraic operations.
8. **Homological algebra:** Studies algebraic structures called homology and cohomology, used in algebraic topology and other areas of mathematics to study geometric objects.

9. Universal algebra: Investigates common algebraic structures and operations across different mathematical systems, providing a unified framework for understanding various algebraic structures.

10. Computational algebra: Focuses on developing algorithms and computational techniques for solving algebraic problems, with applications in cryptography, computer-aided design, and scientific computing.

History of algebra

Algebra has a rich history that stretches back thousands of years. It began with ancient civilizations like the Babylonians and Egyptians, who used algebraic concepts to solve practical problems related to commerce, land measurement, and construction.

The word "algebra" itself comes from the Arabic word "al-jabr," which means "reunion of broken parts." This term was used by the Persian mathematician Muhammad ibn Musa al-Khwarizmi in the title of his book "Kitab al-Jabr wal-Muqabala" (The Book of Restoring and Balancing). Al-Khwarizmi, who lived during the Islamic Golden Age in the 9th century, is often credited as one of the pioneers of algebra.

In his book, al-Khwarizmi introduced systematic methods for solving linear and quadratic equations, laying the foundation for algebra as we know it today. His work was instrumental in the development of algebraic notation and problem-solving techniques.

During the Renaissance in Europe, scholars rediscovered the works of al-Khwarizmi and other Islamic mathematicians, leading to a resurgence of interest in algebra. Mathematicians such as Leonardo Fibonacci and François Viète made significant contributions to the field, further advancing algebraic methods and notation.

In the 16th and 17th centuries, algebra underwent further development with the works of mathematicians like René Descartes and Pierre de Fermat. Descartes introduced the use of coordinate geometry, linking algebra with geometry and paving the way for the development of analytic geometry.

The 18th and 19th centuries saw algebraic concepts applied to various branches of mathematics, including number theory, abstract algebra, and algebraic geometry. Mathematicians such as Carl Friedrich Gauss, Évariste Galois, and Augustin-Louis Cauchy made groundbreaking discoveries that expanded the scope and depth of algebra.

Today, algebra is a fundamental branch of mathematics that plays a central role in various fields, including science, engineering, economics, and computer science. Its rich history reflects the collective efforts of countless mathematicians from diverse cultures and time periods, all contributing to the evolution of algebraic thought and practice.